

A =

$$\begin{pmatrix} x & y & z \\ 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

$$x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

POUŽIJU METODU

NEJMEJŠÍCH ČTVŮRŮ

$$A^T \cdot A \cdot \vec{x} = A^T \vec{z}$$

$$\begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \cdot$$

$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix} = B$$

$$\vec{z} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$$

$$A^T \cdot M_2 = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} =$$

$$= \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} = \vec{u} \quad B \vec{x} = \vec{u}$$

$$\left(\begin{array}{ccc|c} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \end{array} \right) \sim \left(\begin{array}{ccc|c} -1 & 4 & 2 & 4 \\ 4 & -1 & 2 & 2 \\ 2 & 2 & 6 & 6 \end{array} \right) \sim$$

$$\sim \left(\begin{array}{ccc|c} -1 & 4 & 2 & 4 \\ 0 & 15 & 10 & 18 \\ 0 & 10 & 10 & 14 \end{array} \right) \sim \left(\begin{array}{ccc|c} -1 & 4 & 2 & 4 \\ 0 & 10 & 10 & 14 \\ 0 & 15 & 10 & 18 \end{array} \right) \sim$$

$$\sim \left(\begin{array}{ccc|c} -1 & 4 & 2 & 4 \\ 0 & 5 & 5 & 7 \\ 0 & 0 & -5 & -3 \end{array} \right) \sim \left(\begin{array}{ccc|c} -1 & 4 & 2 & 4 \\ 0 & 5 & 5 & 7 \\ 0 & 0 & 1 & \frac{3}{5} \end{array} \right) \sim$$

$$\sim \left(\begin{array}{ccc|c} -1 & 4 & 0 & \frac{14}{5} \\ 0 & 1 & 0 & \frac{4}{5} \\ 0 & 0 & 1 & \frac{3}{5} \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{2}{5} \\ 0 & 1 & 0 & \frac{4}{5} \\ 0 & 0 & 1 & \frac{3}{5} \end{array} \right)$$

$$\vec{x} = \begin{pmatrix} \frac{2}{5} \\ \frac{4}{5} \\ \frac{3}{5} \end{pmatrix}$$